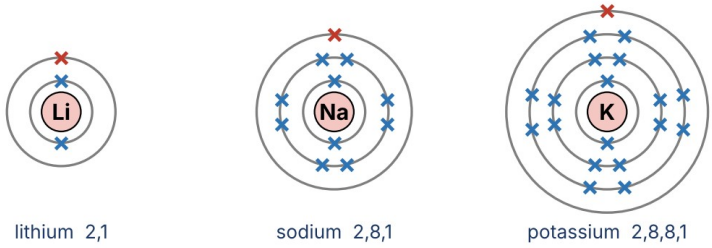
Group 1 – the Alkali Metals

AQA GCSE Combined Science: Trilogy · Chemistry Topic 1 · Revision activity 6 · 19 marks · about 25 minutes

Read the key information (5 min), answer the **quick recall** questions without looking back (5 min), then answer the **exam questions** (15 min). Mark your work with the mark scheme on the last page.

Key information

- The group 1 elements are called the **alkali metals**: lithium, sodium, potassium, rubidium, caesium and francium.
- They all have **one electron in their outer shell**, so they react in similar ways.
- They are **soft** (can be cut with a knife), have **low densities** (Li, Na and K float on water) and are shiny when freshly cut but quickly go dull (tarnish).
- They are very reactive, so they are **stored under oil** to stop them reacting with oxygen and water vapour in the air.
- They react with **water** to form a **metal hydroxide** and **hydrogen**. The solution is **alkaline**.
- They react with **oxygen** to form metal oxides and with **chlorine** to form metal chlorides (white solids).
- **Reactivity increases going down the group.**



lithium 2,1 sodium 2,8,1 potassium 2,8,8,1

Equations

sodium + water → sodium hydroxide + hydrogen $2Na + 2H_2O \rightarrow 2NaOH + H_2$

sodium + oxygen → sodium oxide $4Na + O_2 \rightarrow 2Na_2O$

sodium + chlorine → sodium chloride $2Na + Cl_2 \rightarrow 2NaCl$

Why does reactivity increase down group 1?

Going down the group the atoms have **more shells**, so the outer electron is **further from the nucleus**. There is **weaker attraction** between the nucleus and the outer electron, so it is **lost more easily**.

Reactions with water

Metal	What you see
lithium	floats, fizzes steadily, slowly disappears
sodium	floats, melts into a ball, fizzes quickly, moves around the surface
potassium	floats, melts, fizzes very quickly, hydrogen catches fire with a lilac flame

Section A: Quick recall (8 marks – 1 mark each)

1 What is the name given to the group 1 elements?

2 How many electrons do group 1 atoms have in their outer shell?

3 Why are group 1 metals stored under oil?

4 Name the gas produced when sodium reacts with water.

5 Name the compound in solution after potassium reacts with water.

6 What colour would universal indicator turn in this solution?

7 Which is more reactive: lithium or potassium?

8 Name the product formed when lithium reacts with chlorine.

Section B: Exam questions

01 A small piece of sodium is added to a trough of water. Describe two things you would see.

[2 marks]

02 Write a word equation for the reaction of lithium with water.

[2 marks]

03 Write a balanced symbol equation for the reaction of potassium with water. The formula of potassium hydroxide is KOH.

[2 marks]

04 Explain why potassium is more reactive than sodium.

[3 marks]

05 Rubidium is below potassium in group 1. Predict two things you would observe if rubidium was added to water.

[2 marks]

Mark Scheme – Group 1 – the Alkali Metals

(1) = one mark. / separates acceptable alternatives. Words in brackets are not needed for the mark.

Question	Marking guidance	Comments	Mark
A	**1** the alkali metals **2** 1 **3** to stop them reacting with oxygen / water in the air **4** hydrogen **5** potassium hydroxide **6** purple (or blue) – it is alkaline **7** potassium **8** lithium chloride	1 mark each	8
01	any two from: (1) each • floats • fizzes / bubbles • melts into a ball • moves around the surface • gets smaller / disappears		2
02	lithium + water → lithium hydroxide + hydrogen lithium hydroxide (1) hydrogen (1)		2
03	$K + H_2O \rightarrow KOH + H_2$ – correct formulae (1) $2K + 2H_2O \rightarrow 2KOH + H_2$ – balanced (1)		2
04	potassium has more shells / its outer electron is further from the nucleus (1) so there is weaker attraction between the nucleus and the outer electron (1) so the outer electron is lost more easily (1)		3
05	any two from: (1) each • reacts more violently than potassium / may explode • fizzes very rapidly • melts / catches fire • solution becomes alkaline (indicator turns purple)		2

Total: 19 marks